MODEL LAB · 2026-10-24

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

20,133
held-out plate appearances

2.97%
held-out HR rate

0.6009
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-10, scored on the 20,133 that came after (actual HR rate 2.97%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5832	1.51×	0.02872	0.13267	0.00563
KNN comps Prior is the pooled rate of the nearest comparables.	0.6009	1.63×	0.02868	0.13158	0.00464
SVR Support-vector regression on the same feature matrix.	0.5952	1.44×	0.02871	0.13232	0.00600
Random forest Forest regression; a learned similarity metric.	0.5980	1.54×	0.02869	0.13175	0.00464
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6009	1.63×	0.02868	0.13158	0.00464
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6009	1.63×	0.02868	0.13158	0.00464
Linear regression Ordinary least squares on the same features.	0.5974	1.57×	0.02869	0.13193	0.00514
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5975	1.63×	0.02871	0.13204	0.00552
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5976	1.63×	0.02869	0.13188	0.00543
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5960	1.54×	0.02870	0.13212	0.00552

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **257** knn **257** svr **257** rf **257** core **257** ensemble **257** linear **257** logistic **257** nn **257** xgb **257**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (1,884 PA, actual HR rate 2.44%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.6009	1.31×	0.02383	0.11484	0.01705
KNN comps Prior is the pooled rate of the nearest comparables.	0.6332	2.18×	0.02370	0.11208	0.01018
SVR Support-vector regression on the same feature matrix.	0.6097	1.52×	0.02393	0.11661	0.01831
Random forest Forest regression; a learned similarity metric.	0.6353	1.96×	0.02372	0.11209	0.01174
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6332	2.18×	0.02370	0.11208	0.01018
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6332	2.18×	0.02370	0.11208	0.01018
Linear regression Ordinary least squares on the same features.	0.6390	2.18×	0.02375	0.11327	0.01360
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6346	1.96×	0.02380	0.11321	0.01386
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6312	2.18×	0.02372	0.11258	0.01436
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6196	1.31×	0.02388	0.11478	0.01757

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Junior Caminero 15.3%, Jackson Chourio 14.9%, Jake Bauers 14.7%, Jonathan Aranda 14.6%, William Contreras 13.2%, Yandy Díaz 12.8%
Flat league	6/6	Jackson Chourio 15.1%, Junior Caminero 14.7%, Jake Bauers 13.9%, Jonathan Aranda 13.9%, William Contreras 13.1%, Yandy Díaz 13.1%
KNN comps	6/6	Jackson Chourio 15.4%, Jake Bauers 15.2%, Junior Caminero 15.1%, Jonathan Aranda 14.6%, William Contreras 13.2%, Yandy Díaz 12.9%
SVR	6/6	Junior Caminero 16.5%, Jackson Chourio 15.8%, Jake Bauers 15.0%, Jonathan Aranda 14.4%, Yandy Díaz 13.4%, William Contreras 13.2%
Random forest	6/6	Jackson Chourio 15.7%, Junior Caminero 15.5%, Jake Bauers 15.3%, Jonathan Aranda 14.5%, Yandy Díaz 13.1%, William Contreras 13.1%
Core (KNN+SVR)	6/6	Jackson Chourio 15.4%, Jake Bauers 15.2%, Junior Caminero 15.1%, Jonathan Aranda 14.6%, William Contreras 13.2%, Yandy Díaz 12.9%
Forest blend	6/6	Jackson Chourio 15.4%, Jake Bauers 15.2%, Junior Caminero 15.1%, Jonathan Aranda 14.6%, William Contreras 13.2%, Yandy Díaz 12.9%
Linear regression	6/6	Jackson Chourio 15.5%, Junior Caminero 15.4%, Jake Bauers 14.9%, Jonathan Aranda 14.6%, William Contreras 13.2%, Yandy Díaz 13.0%
Logistic regression	6/6	Junior Caminero 15.8%, Jackson Chourio 15.6%, Jake Bauers 14.9%, Jonathan Aranda 14.3%, William Contreras 13.2%, Yandy Díaz 12.9%
Neural net (torch)	6/6	Junior Caminero 15.8%, Jackson Chourio 15.6%, Jake Bauers 15.1%, Jonathan Aranda 14.4%, William Contreras 13.2%, Yandy Díaz 12.9%
XGBoost	6/6	Junior Caminero 16.1%, Jackson Chourio 15.6%, Jake Bauers 15.2%, Jonathan Aranda 14.6%, Yandy Díaz 13.3%, William Contreras 13.0%

Hitter by hitter

The top 40 bats by the served probability.

	HITTER		SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1	**Junior Caminero**	TB	**15.3%**	14.7%	15.1%	16.5%	15.5%	15.1%	15.1%	15.4%	15.8%	
2	**Jackson Chourio**	MIL	**14.9%**	15.1%	15.4%	15.8%	15.7%	15.4%	15.4%	15.5%	15.6%	
3	**Jake Bauers**	MIL	**14.7%**	13.9%	15.2%	15.0%	15.3%	15.2%	15.2%	14.9%	14.9%	
4	**Jonathan Aranda**	TB	**14.6%**	13.9%	14.6%	14.4%	14.5%	14.6%	14.6%	14.6%	14.3%	
5	**William Contreras**	MIL	**13.2%**	13.1%	13.2%	13.2%	13.1%	13.2%	13.2%	13.2%	13.2%	
6	**Yandy Díaz**	TB	**12.8%**	13.1%	12.9%	13.4%	13.1%	12.9%	12.9%	13.0%	12.9%	
7	**Christian Yelich**	MIL	**12.3%**	12.5%	12.6%	11.9%	12.5%	12.6%	12.6%	12.7%	12.1%	
8	**Garrett Mitchell**	MIL	**11.6%**	11.3%	11.7%	10.6%	11.3%	11.7%	11.7%	11.3%	11.2%	
9	**Brice Turang**	MIL	**11.4%**	11.6%	11.1%	11.2%	11.0%	11.1%	11.1%	10.8%	10.7%	
10	**Victor Mesa Jr.**	TB	**10.7%**	10.6%	10.8%	11.3%	11.2%	10.8%	10.8%	11.0%	11.0%	
11	**Richie Palacios**	TB	**8.2%**	9.1%	8.0%	7.9%	7.4%	8.0%	8.0%	7.8%	7.9%	
12	**Liam Hicks**	TB	**7.7%**	8.9%	7.9%	8.4%	8.1%	7.9%	7.9%	8.1%	8.0%	
13	**Jonny DeLuca**	TB	**7.5%**	8.0%	6.9%	7.1%	6.8%	6.9%	6.9%	7.1%	7.4%	
14	**David Hamilton**	MIL	**6.4%**	7.8%	6.3%	6.0%	5.6%	6.3%	6.3%	6.3%	6.2%	
15	**Sal Frelick**	MIL	**5.9%**	7.5%	6.1%	5.4%	5.5%	6.1%	6.1%	5.8%	5.9%	
16	**Cooper Pratt**	MIL	**5.7%**	7.4%	5.7%	5.2%	5.4%	5.7%	5.7%	4.9%	5.1%	
17	**Taylor Walls**	TB	**5.1%**	6.6%	5.2%	3.9%	3.9%	5.2%	5.2%	4.9%	4.9%	
18	**Chandler Simpson**	TB	**3.3%**	7.0%	4.6%	1.9%	2.0%	4.6%	4.6%	1.9%	3.8%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

Slate for 2026-10-24 · model 2026.10.01-stack · built 2026-10-25 04:33:40 · live data 2026-10-25 04:33:41

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